Build the berth

Purpose, evidence and mission contract

SALTLINE is a single-player industrial-port FPS. Cut three uplink relays, survive the responding machines, and reach the boat. Rain, salt haze and wet steel frame a compact berth where exposed lanes alternate with container cover and warehouse shadow. The intended rhythm is reconnaissance, a committed relay hold, a sharp reinforcement response, then relocation.

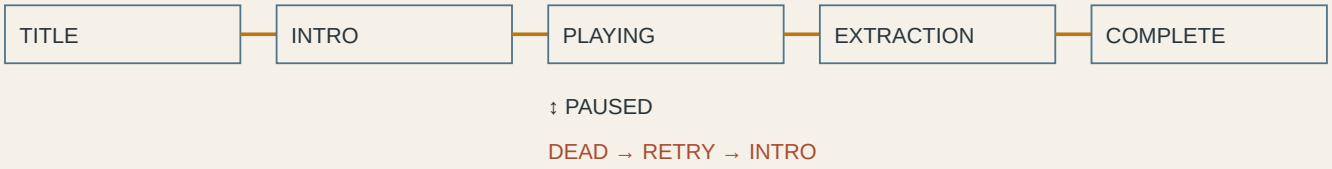


Figure 1: Mission sequence. Extraction is a condition inside PLAYING, part of the active mission.

Design and asset package

This document specifies the playable mission in metres and seconds. The accompanying `gallery/assets/*.js` files define executable model construction; retain their proportions, surfaces and articulation when translating them to another engine. The gallery provides orbitable inspection views. The following chapters and detailed rule sheets define movement, combat, enemy behavior and level placement independently of the asset code.

Mission contract

Deploy at the gate. All three relays begin live and may be cut in any order. Hold interaction while near and facing a live relay. Cutting a relay schedules reinforcements. After the third cut, markers change to extraction and pursuit continues.

Entering extraction while alive and after all cuts completes the run. Being killed opens the death flow; retry resets the mission and increments retries. Pause suspends gameplay; losing pointer lock after acquisition or hiding the page also pauses.

Quantitative rules

Parameter	Value
introSeconds	3.5
retryIntroSeconds	1.3
interactRange	2.2
interactFacing	0.45
cutSeconds	2.4
maxAlive	9
spawnStagger	0.85
spawnFar	18

Distances use world metres; time uses seconds. Facing is a dot-product threshold. Relay selection also rejects vertical separation above 2.5 m; very close targets bypass the facing test.

Completion and scoring

Extraction requires horizontal distance below 2.5 m and vertical difference below 3.5 m. A run starts at full health and resets weapons, effects, combat counters and relay states.

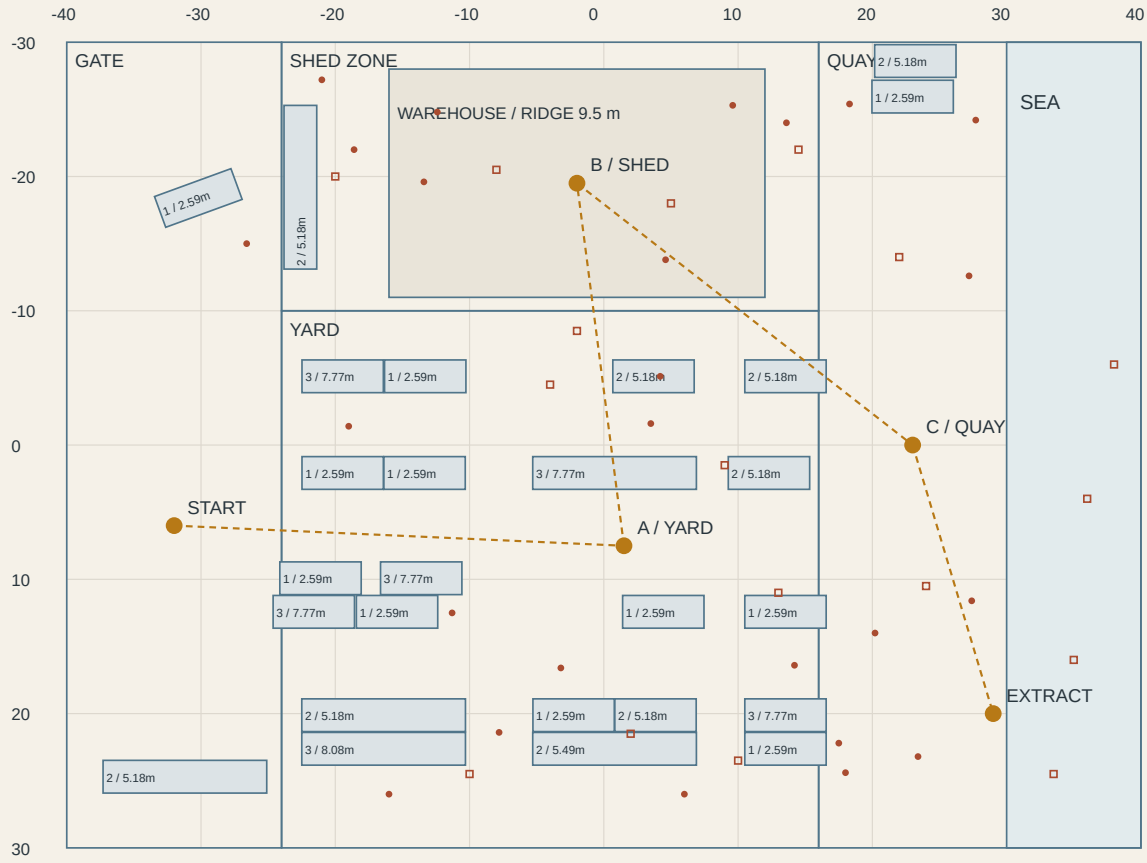
The debrief combines time, accuracy and retries: time below 270 s earns two points, below 400 s one; accuracy at least 50% earns two, at least 30% one; zero retries adds one. Ratings span Messy, Loud, Steady, Clean, Sharp and Surgical.

Reconstruction priorities

Preserve traversable scale, target readability, enemy warning time, weapon cadence and mission pacing. Small bevels, fastener counts and cosmetic noise can be interpreted after those contracts work.

The playable yard

Coordinate map / mission locations / structure footprints



X → east / Z ↓ south / metres / Y is elevation

● Ground spawn □ Air spawn ● Amber: mission location

Container labels: tier count / top height. See placement schedules for offsets and dressing.

Dashed connections explain mission flow; they are not verified walkable routes.

Figure 2: Top-down placement guide. Spawn dots are authored candidates; runtime navigation repair and jitter may move ground spawns.

Coordinates and scale

Y is vertical. Map X increases right and Z down. The player camera starts facing −Z before yaw; the robot builders use +Z as forward. Preserve that distinction when converting orientations.

The warehouse spans X −16…12 and Z −28…−11; floor Y 0.03, eaves 7 m and ridge 9.5 m. The quay edge is X 30. The ladder and boat extraction align with Z 20.

Mission positions

Location	X / Y / Z
player	−32 / 0 / 6
relay_a	1.5 / 0 / 7.5
relay_b	−2 / 0.03 / −19.5
relay_c	23 / 0 / 0
extraction	29 / 0 / 20

The drawn links express sequence only. Players can choose relay order; containers and walls determine the actual routes.

Space, cover & collision

Level assembly / navigation / encounter placement



WAREHOUSE / Z–Y SECTION

Depth: 17 m / X span: 28 m

Eaves: 7 m / ridge: 9.5 m

Floor: Y 0.03 m / roof ridge: Z −19.5

Structural envelope from no; doors, skylights, rafters and interior props omitted.

Figure 3: Source-dimension warehouse section; structural envelope only.

Build the space in layers

Start with the ground plane, warehouse shell, quay and world boundaries. Add container stacks, stairs, catwalks and major cover. Then place relays and validate access before adding dressing such as pallets, cables, bollards and decals.

The drawing uses first-tier container sizes from the prop registry. The placement schedule later in this document specifies every container tier, including offsets, rotations and high-cube heights.

Authored zones

Zone	X range	Z range
Gate	−40…−24	−30…30
Yard	−24…16	−10…30
Shed	−24…16	−30…−10
Quay	16…40	−30…30

Zone rectangles classify gameplay space. They are not complete physical boundary walls, and the shed zone is larger than the warehouse footprint.

Collision contracts

The game uses box colliders, capsule movement, raycasts, ground-height queries and a navigation grid. Collider flags distinguish navigation exclusions, bullet-only geometry and player-only blockers. Keep these meanings separate when rebuilding collision in an engine.

Raycasts serve shooting and sensing; ground queries serve foot placement and movement. Visible geometry and collision need not have identical detail. Thin railings, decals and cables should not accidentally seal a walking route.

Vertical routes

A yard vantage includes stair-foot location (−2.2, 0, 12.4) and roof location (13.5, 2.59, 12.4). The catwalk connects elevated cover near the container row. Verify stair approach and overhead clearance with the player capsule.

The quay contains a ladder opening and boat destination. Falling 2.2 m below sea level returns the player to the last grounded safe position inland of X 28.8, zeros velocity, turns yaw by π and deals 12 damage. Landing after more than 0.12 s airborne deals 7 × (impact speed − 13) damage when impact speed exceeds 13 m/s.

Spawn selection

The level supplies 42 candidate spawns. Ground spawns use the navigation grid; air spawns retain altitude. Candidates are scored for distance, concealment and recent use. A preferred spawn distance of 18 m is a scoring preference rather than an absolute rejection boundary.

Recent positions are penalized for six seconds. Spawn jitter is up to ±1.2 m in X/Z and ground jitter requires walkability. Candidate repair searches neighboring walkable cells before gameplay.

Cover and sight

Authored cover points support the container lanes and quay. Hounds navigate toward firing positions; gulls move in three dimensions. Validate sight lines from player eye height and enemy sensors, not only from ground-plane points.

Implementation acceptance: every relay has an accessible approach, all starting paths fit the capsule, the elevated route is usable, and extraction remains reachable after reinforcements appear.

Player & camera

Movement tuning / bodily feedback / interaction

Movement contract

The player is a capsule with radius 0.32 m and standing height 1.8 m. Standing eye height is 1.68 m; crouched eye height is 1.05 m. Keep collision height and camera height distinct.

Parameter	Value
walk	3.8
sprint	6.4
crouch	1.9
ads	2.4
accelGround	34
decelGround	26
accelAir	9
gravity	21
jumpApex	1
coyote	0.12
jumpBuffer	0.13
crouchHeight	1.15

Speeds are m/s; accelerations and gravity are m/s². Jump apex is 1 m; launch velocity is derived from gravity and apex. Coyote time and input buffering make jumps tolerant around ledges and landing.

Camera contract

World FOV defaults to 74 degrees, with a separate 54-degree viewmodel camera. Sprint adds 6 degrees; aimed FOV targets 52 degrees. Pitch is limited to ±1.5 radians.

Aimed look scales by 0.6. Camera movement includes stride-based bob, strafe roll, landing dip, recoil recovery and damage shake. These are presentation effects; avoid letting them unpredictably move the gameplay collision shape.

Health and damage

The player starts at 100 health. Damage resets the regeneration timer; regeneration begins after 4 s and restores 25 health/s. Death occurs at zero. Hits trigger directional feedback and camera shake.

Low health causes visual feedback and a one-time radio line below 30 health. Provide cover intervals long enough for the regeneration rule to matter.

Relay hold interaction

A live relay must be within 2.2 m, within vertical tolerance and sufficiently in front. Holding interaction advances normalized progress by elapsed time divided by 2.4. Completing it permanently cuts that relay for the run.

Leaving the interaction or releasing input decays accumulated progress at 1.6 normalized units/s. It does not instantly reset the internal progress to zero. The relay's visual state returns to live while interrupted.

Input and pause

Support movement, look, fire, aim, sprint, crouch, jump, reload, interaction and pause as actions. The game has keyboard/mouse, gamepad and touch paths, with pointer-lock and drag-look modes. Rebinding is an implementation choice, not a claim about the game UI.

Acceptance: walking the same path at different frame rates produces consistent speed; the player cannot stand into a ceiling; interrupted interaction decays; pause stops gameplay and resume restores input.

PAC-9 weapon system

Active viewmodel / firing / reload / impact pipeline



Figure 4: PAC-9 side view rendered from gallery/assets/pac9.js. Hands and transient effects are excluded.

PAC-9 specification

Rule	Value
Magazine	30 rounds
Reserve	150 rounds
Cadence	720 rpm
Shot damage	26
Range	300 m
Tactical reload	2.1 s
Empty reload	2.72 s

At 720 rpm, the nominal shot interval is 0.0833 s. Treat timer accumulation carefully so low frame rates do not silently reduce cadence.

Visual assembly

Use separate receiver, barrel/handguard, grip, magazine, bolt and optic parts. The game separates anodized metal, phosphate metal, polymer, tan polymer, brass, glove fabric and rubber. Hands follow the weapon pose with dedicated finger and thumb geometry.

The viewmodel uses its own camera and suppresses world shadow casting. Aim alignment moves the weapon to the sight line rather than simply zooming the world camera.

Fire pipeline

Input and cooldown decide whether a shot can occur. The weapon consumes ammunition, creates a shot direction including spread, and sends it through combat. Combat resolves enemy or world hits, then drives hitmarkers, effects and sound.

The weapon also animates recoil, bolt travel, muzzle flash, shell ejection, heat and smoke. Shots originate at the player eye; muzzle-to-impact tracers are cosmetic. Muzzle visuals and the gameplay ray have different roles.

Reload timeline

A reload tracks normalized time, empty/tactical status and phase. The game drives magazine removal, replacement and bolt actions from this sequence. Reload audio should follow those milestones, not a single unchanging sound.

Magazine removal occurs at 0.47 s; the replacement becomes visible after 0.86 s; insertion transfers min(magazine capacity − rounds remaining, reserve) at 1.345 s. A seating tap occurs at 1.5 s. Empty reloads pull the bolt from 2.02 to 2.14 s, hold until 2.19 s and release over 0.035 s. Fire is blocked throughout the reload. A dry trigger clicks once per press and schedules reload after 0.32 s when reserve exists.

Aim blends over 0.18 s with quintic easing. Full spread and recoil rules appear in the detailed combat sheet.

Feedback priorities

A hit must be distinguishable from a miss; a kill must be distinguishable from an ordinary hit. Surface impacts vary between metal, concrete, wood, water and glass. Reserve and magazine values must remain synchronized with the HUD.

Implementation scenarios

Verify tactical reload with a partly full magazine, empty reload with reserve available, reserve exhaustion, dry firing, holding fire, aim transitions, sprint interruption, pause during reload and retry reset. Confirm that cosmetic recoil and muzzle flash do not introduce duplicate damage.

Interpretation: use a single authoritative shot result to drive damage, sound, particles and UI. This preserves consistent feedback even if an engine uses different rendering and animation APIs.

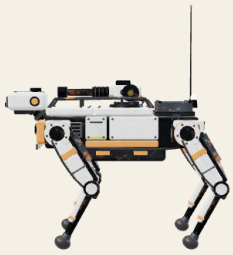
Hound / construction

H-03 / narrow chassis / optical head / dorsal gun



Figure 5: Original hound geometry and procedural surfaces, rendered from gallery/assets/hound.js. Static foot placement uses the original IK solver.





Proportional anchors

Parameter	Value
stand	0.72
L1	0.4
L2	0.42
l0	0.075
hipX	0.165
hipY	-0.01
hipZ	0.4
footR	0.046

Main shell deck width is 0.35 m; the metal body is 0.27 × 0.20 × 0.98 m. The head envelope is 0.19 × 0.118 × 0.25 m. The long antenna and projecting gun are important silhouette cues.

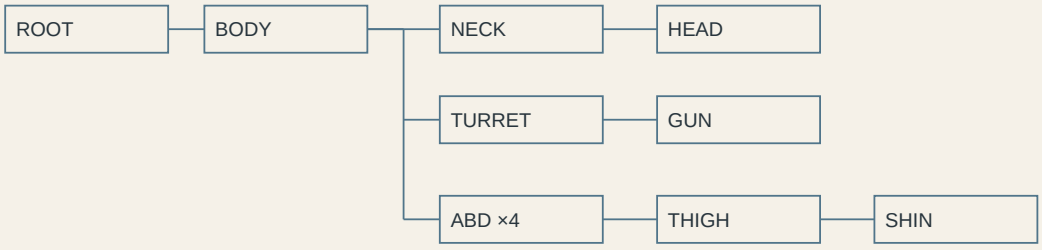
Assembly reading

The extracted JavaScript builds the original skinned geometry and its joint hierarchy. White shell panels sit over dark structural metal, with orange bands and bright optics. Small screws, handles, fins and cables give scale but are secondary to chassis width, leg spacing and head position.

Edit `gallery/assets/hound.js` to change parts and local transforms. The gallery and PNG renderer execute that file directly; no JSON geometry conversion sits between source and image. Shared primitive and skinning helpers remain in `gallery/shared/reference.js`.

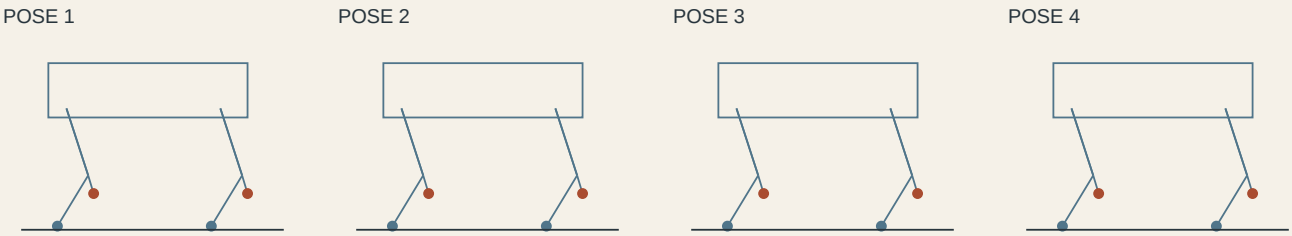
Hound / motion & threat

Rig relationships / planted feet / attack communication



Abduction Z • thigh/knee X • turret Y • gun X • separate antenna and beacon children

Figure 6: Logical articulation tree; four legs repeat the abduction → thigh → shin chain.



Schematic diagonal stepping: FL + RR alternate with FR + RL. Swing arcs lift; stance feet stay planted.

Figure 7: Movement explanation, not extracted animation keyframes. The actual rig solves joints from world-space foot targets.

Grounded movement

The game alternates diagonal pairs FL+RR and FR+RL. Stance feet retain world positions. Swing targets anticipate movement and turning; a smooth interpolation moves the foot while a sine arc lifts it above the ground. Ground-height queries set landing height.

Inverse kinematics solves hip abduction, thigh and knee angles. Body height, pitch and roll respond to the four foot heights, acceleration and turning. Springs soften those responses. The head aims independently, the turret yaws, the gun pitches and the antenna sways.

This is substantially richer than swinging every limb with a sine wave. Preserve planted feet before adding secondary motion.

Threat values

Parameter	Value
hp	140
shotDamage	5
burst	4
burstGap	0.11
telegraph	0.5
runSpeed	4.5
patrolSpeed	1.5
lungeRange	3
lungeDamage	15
lungeCooldown	3.5
lungeSpeed	9.5

Behavior and warnings

Hounds patrol, investigate, search and engage. Their combat sequence includes telegraph, burst fire, cooldown and close-range lunge. Head and lens feedback make the attack legible. Weapon aim is not the same as chassis heading.

Hounds smoke below 56 health (40% of maximum). Death changes body posture, releases attack ownership, slides feet and creates a wreck. The chassis drops to 0.215 m with a 0.22 bounce factor; legs splay and slide, head droops and the optics flicker for 1.2 s. Smoke lasts 6.5 s. Retain five wrecks; older wrecks sink at 0.22 m/s for 3.2 s before removal.

Gull & encounter pacing

Airborne enemy / shared senses / relay reinforcement



Figure 8: Original gull geometry rendered from gallery/assets/gull.js. Motion-blur discs and the searchlight cone are omitted for this static view.

Gull construction

Four rotor centers are arranged at X ±0.33 and Z ±0.30 m. The body carries rotor supports and protective rings, landing skids, navigation strobes, an underslung sensor gimbal, lamp and gun. Rotor motion and transparent rotor discs serve different visual roles.

The lamp/sensor ball pitches below a yaw pivot. The gun pitches separately. Keep the sensor line of sight and muzzle position tied to their actual transforms.

Parameter	Value
hp	45
boltDamage	7
boltSpeed	70
burst	3
burstGap	0.12
telegraph	0.55
rangeMin	12
rangeMax	25
altMin	4
altMax	8
maxSpeed	8
fireRange	42

Flight and destruction

Gulls orbit during engagement, investigate last-known positions and avoid obstructions with directional feelers. They check clearance above and below, steer with acceleration and bank in response to forces. They do not follow the hound navigation grid.

At death, a rotor is lost, gravity and spin take over, smoke appears, and collision ends in an explosion. Explosion radius is 3 m; damage is 30.

Shared AI contracts

Sight, hearing and ranged attack ownership control combat pressure.

Sight uses a 45 m range and 200° horizontal cone, bypassed within 3.5 m or while engaged. Sample visibility every 0.11–0.19 s. Hounds require 0.55 s visible contact; gulls 0.4 s; investigating/searching enemies need 45% of that. Loss of sight for over 4 s enters an 8–11 s search, then patrol. Sight accumulates exposure at 0.1/s and absence removes 0.45/s.

Two enemies may own ranged attacks simultaneously, rising to three above five living enemies. Grants are at least 1.25 s apart. If the other species requested in the past 0.5 s, a repeated-species grant waits until 2.75 s after the previous grant. Release ownership after bursts, cancellation, death or removal. Melee does not require ownership.

An alert rallies non-engaged allies within 18 m to investigate for 9–13 s. Taking a hit provokes investigation for 10 s with immediate sensing and rallies within 20 m. Gunshots alert within 45 m, throttled to once per 0.3 s; explosions alert within 30 m. Hearing supplies a location rather than immediate permission to shoot.

Reinforcement schedule

Trigger	Gulls	Hounds
Initial patrol	2	1
Cut 1	3	1
Cut 2	3	2
Cut 3	4	3

Waves depend on cut count, not relay identity. The first scheduled spawn is 1.6 s after the cut; subsequent entries are staggered by 0.85 s. The active cap is 9 queued entries defer when capped.

After all cuts, pursuit starts with a 12 s timer and up to 5 additional enemies. The nominal pursuit interval is 7 s, gated by queue and alive-count conditions. The game also periodically alerts survivors toward the player.

Surface & atmosphere

Materials / weather / rendering / quality tiers

Material vocabulary

The game creates procedural albedo, normal and packed surface textures. Named materials provide painted steel, concrete, asphalt, brick, rubber, polymer, fabric, wood, water and glass. Decals add labels, hazards, grime, rust runs, bullet marks and scorch.

Hound geometry groups into shell, metal, glow and lens slots. Its vertex colors include values above one, so these are not plain display-sRGB paint chips. Preserve the intended light/dark hierarchy rather than interpreting raw triples as final screen colors.

Family	Visual intent
Shell	Light painted armor; orange bands
Metal	Dark structure; bright machined fasteners
Rubber	Near-black hoses and feet
Lens / glow	Amber idle; redder attack feedback
Ground	Wet asphalt and concrete; puddle variation
Sea	Dark moving surface with horizon haze

Readability in weather

Wetness changes roughness and surface response. Rain includes falling streaks and splashes; interior bounds suppress outdoor effects. Puddles and the sea use different scales. Do not apply kilometre-scale sea texture tiling to small puddles.

Lighting and post-processing

The renderer includes sky and sun lighting, shadows, fog/volumetrics, ambient occlusion, bloom, adaptation, lens dirt, streaks, flare, vignette, grain and sharpening. These effects support the industrial atmosphere but must leave relay lights and attack cues readable.

Damage, low health, aiming, flash, fade and saturation are exposed as post-process signals. Drive them from gameplay events rather than mixing game rules into a shader.

Quality tiers

Tier	Scale	Shadow
low	0.7	2048
medium	0.85	2048
high	1	4096
ultra	1	4096

Quality changes may reduce MSAA, texture resolution, volumetric steps, ambient occlusion, planar reflections, bloom, motion blur and rain density. Runtime scaling can modify effective resolution.

Reconstruction guidance

Build a readable daylight scene first, then add sky/fog and wet surfaces. Validate combat cues before enabling glare and grain. Quality reductions should change visual cost while preserving collision, sensing, interaction reach and weapon behavior.

Acceptance: optics remain visible against both warehouse shadow and bright sky; rain does not visually continue through the warehouse; texture repetition remains plausible at player scale.

Asset thumbnails / 1

Executable JavaScript builders / original procedural surfaces / studio lighting



container20
6.07 × 2.59 × 2.45 m / nominal
assets/containers.js



container40
12.2 × 2.59 × 2.44 m / nominal
assets/containers.js



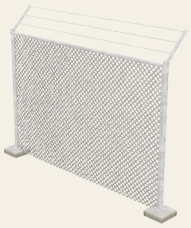
container40hc
12.2 × 2.9 × 2.44 m / nominal
assets/containers.js



jersey barrier
3 × 0.86 × 0.61 m / nominal
assets/jersey_barrier.js



light mast
0.99 × 14 × 1.9 m / nominal
assets/light_mast.js



fence panel
3.32 × 2.4 × 0.53 m / nominal
assets/fence_panel.js



guard booth
3.11 × 2.55 × 2.77 m / nominal
assets/guard_booth.js



gate arm
5.34 × 1.19 × 0.46 m / nominal
assets/gate_arm.js



bollard mooring
0.74 × 0.61 × 0.74 m / nominal
assets/bollard_mooring.js

Source of truth

Every tile is a real Three.js render of its named builder. Edit the listed file under `gallery/`, then rebuild to refresh both the gallery and PDF. Nominal dimensions are read from the extracted JavaScript registry.

Assets are framed individually, so image sizes are not mutually to scale. Lighting and exposure are standardized; construction geometry and surfaces come from the executable builders. Detailed colliders remain in the builders.

Reading the views

Asset thumbnails / 2

Executable JavaScript builders / original procedural surfaces / studio lighting



bollard post
0.17 × 1 × 0.18 m / nominal
assets/bollard_post.js



relay node
2.57 × 3.18 × 1.17 m / nominal
assets/relay_node.js



shelf rack
2.75 × 4.2 × 1.13 m / nominal
assets/shelf_rack.js



stairs metal
3.12 × 3.62 × 1.08 m / nominal
assets/stairs_metal.js



catwalk
3 × 1.25 × 1.22 m / nominal
assets/catwalk.js



reefer unit
0.63 × 2.3 × 2.3 m / nominal
assets/reefer_unit.js



barrel
0.59 × 0.9 × 0.59 m / nominal
assets/barrel.js



pallet
1.21 × 0.14 × 1.01 m / nominal
assets/pallet.js



pallet stack
1.26 × 1.15 × 1.05 m / nominal
assets/pallet_stack.js

Source of truth

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Reading the views

Asset thumbnails / 3

Executable JavaScript builders / original procedural surfaces / studio lighting



crate wood
1.11 × 1 × 0.91 m / nominal
assets/crate_wood.js



crate stack
1.21 × 1.91 × 1.04 m / nominal
assets/crate_stack.js



ibc tote
1.2 × 1.14 × 1 m / nominal
assets/ibc_tote.js



forklift
3.41 × 2.19 × 1.25 m / nominal
assets/forklift.js



cable reel
1.3 × 1.3 × 0.86 m / nominal
assets/cable_reel.js



tire stack
0.81 × 0.99 × 0.81 m / nominal
assets/tire_stack.js



dumpster
1.92 × 1.21 × 1.22 m / nominal
assets/dumpster.js



generator
2.22 × 1.41 × 1.04 m / nominal
assets/generator.js



jerry can
0.35 × 0.45 × 0.17 m / nominal
assets/jerry_can.js

Source of truth

Every tile is a real Three.js render of its named builder. Edit the listed file under `gallery/`, then rebuild to refresh both the gallery and PDF. Nominal dimensions are read from the extracted JavaScript registry.

Assets are framed individually, so image sizes are not mutually to scale. Lighting and exposure are standardized; construction geometry and surfaces come from the executable builders. Detailed colliders remain in the builders.

Reading the views

Asset thumbnails / 4

Executable JavaScript builders / original procedural surfaces / studio lighting



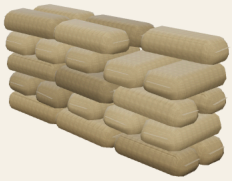
traffic cone
0.36 × 0.7 × 0.36 m / nominal
assets/traffic_cone.js



gas cylinders
0.91 × 1.6 × 0.5 m / nominal
assets/gas_cylinders.js



tarp cargo
2.41 × 1.46 × 1.48 m / nominal
assets/tarp_cargo.js



sandbags
1.99 × 0.9 × 0.62 m / nominal
assets/sandbags.js



toolbox cart
0.86 × 0.96 × 0.54 m / nominal
assets/toolbox_cart.js



floodlight stand
0.98 × 2.24 × 1.65 m / nominal
assets/floodlight_stand.js



cardboard boxes
0.98 × 0.74 × 0.92 m / nominal
assets/cardboard_boxes.js



pipe stack
5.96 × 0.84 × 1.2 m / nominal
assets/pipe_stack.js



sign post
0.6 × 2.41 × 0.3 m / nominal
assets/sign_post.js

Source of truth

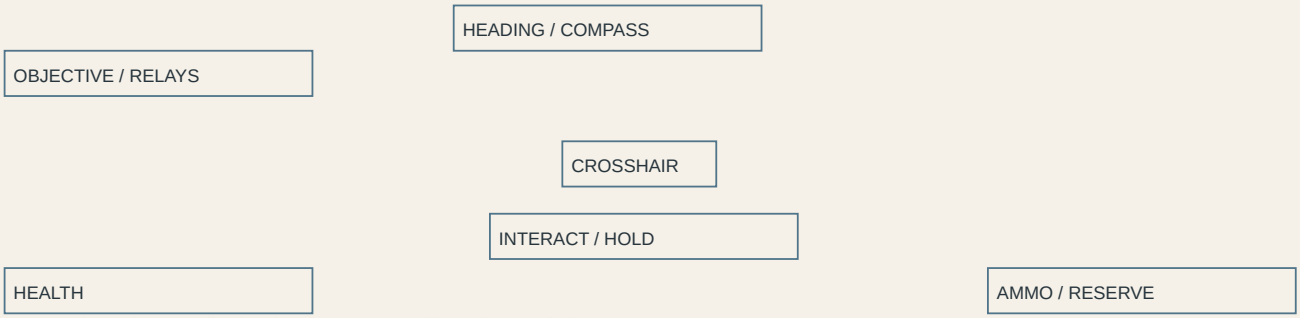
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Reading the views

Assets are framed individually, so image sizes are not mutually to scale. Lighting and exposure are standardized; construction geometry and surfaces come from the executable builders. Detailed colliders remain in the builders.

HUD, audio & delivery

Presentation contracts / build instructions / implementation checks



Layout schematic; normalized to the reference HUD hierarchy, not pixel-exact.

Figure 9: Screen-space hierarchy of the game HUD. Exact spacing adapts to display and input mode.

HUD and menus

Display heading, relay objective/count, world markers, crosshair, hitmarkers, directional damage, interaction progress, radio messages, health, magazine, reserve and reload cues. Pause, death and completion have distinct screens.

Settings include quality, sensitivity, invert-Y, volume and FOV. Touch mode changes layout and supplies stick/fire/aim/jump/reload/interact/pause controls. Keep control prompts synchronized with the active input mode.

Sound event map

Event	Sound family
Player fires	Rifle shot and interior/outdoor tail
Reload phase	Magazine out / in / bolt
Footstep	Concrete / water / metal
Enemy	Rotor / servo loops, warning, firing, death
World	Rain, sea, wind, thunder, foghorn
Mission	Relay cut, objective, completion, death

The game synthesizes audio buffers and spatializes world sounds. Music intensity changes from quiet patrol to contact and active waves/extraction. Interior transitions affect ambience and reverb. Pause and pooled-enemy cleanup must stop or attenuate the appropriate loops.

Verification checklist

Validate movement and capsule clearances; all relay approaches; interruption decay; wave cap and retry cleanup; tactical/empty reload; enemy warning cancellation; grounded feet; air clearance; extraction gating; sound loop retirement; and HUD consistency.

The document build checks extraction counts, SVG XML validity and successful Typst compilation. It does not prove game equivalence. Compare a reconstructed level and enemy behavior against the original in play.

Reading the catalogue

Studio thumbnails use dry surfaces and fixed poses to clarify construction. In play, use the wet lighting and moving articulation specified here. Individually framed images do not share an image scale; use their metre dimensions.

Combat rulebook

Spread, recoil and enemy attack state machines

Shot distribution and recovery

Let A be eased aim blend, S = clamp(horizontal speed / 6, 0, 1), and B be bloom. Target angular spread in radians is:

```
lerp(.0125,.0012,A)    +    S*lerp(.013,.004,A)    +
B*lerp(.0065,.0022,A)     +     airborne*.022     -
crouched*.002*(1-A).
```

Follow this target exponentially at rate 18/s. Sample angle uniformly over 2π and radius as tan(spread) × sqrt(uniform random); normalize (cos(angle) × radius, sin(angle) × radius, −1) in camera space. Each shot adds 0.17 bloom, capped at 1; bloom falls by 1.5/s. Aim uses a linear 0.18 s transition eased with `6t^5 - 15t^4 + 10t^3`.

Held fire catches up missed shot intervals, at most three shots per update. Cooldown cannot accumulate more than 0.001 s credit while idle. Fire requires no reload and sprint blend below 0.45; aim requires sprint blend below 0.35. Sprint blend follows at exponential rate 9/s and is suppressed by fire, aim or reload. Burst index resets after 0.24 s without shooting, randomizing lateral phase.

Recoil

Per-shot multiplier M = lerp(1, .55, A) × (crouched ? .85 : 1). Pitch impulse = .0068 × M × uniform(.85,1.15), reduced to 38% after the ninth consecutive shot. For shots 1–3 yaw is uniform(−.0011,.0011) × M; subsequently yaw is `(sin(shotIndex*.57 + phase)*.0042 + uniform(-.001,.001))*M`.

Keep 35% of pitch/yaw permanently in look orientation. The remaining 65% is recoverable displacement, clamped to ±0.3 pitch and ±0.2 yaw. Hold recovery for .09 s; then decay at 6.5/s. Displayed displacement follows at 34/s.

Viewmodel recoil uses damped springs: backward translation stiffness/damping 520/30, pitch 380/26, yaw 300/22, roll 300/20. Shot velocity impulses are respectively 1.5 × lerp(1,.6,A), 1.5M, uniform(−.5,.5)M, and uniform(−.9,.9)M. Use spring acceleration = −stiffness × displacement − damping × velocity. Heat rises .05/shot to 1.3 and cools .2/s. Above .4 heat, stopping fire produces muzzle smoke after .11 s.

Hound attack machine

Engagement begins by selecting a firing position. Move at 4.5 m/s; reconsider when the player moves over 7 m. Aim on arrival, after 7 s moving, or probabilistically after 1.4 s with visibility and range 7–20 m (probability .7 × elapsed seconds per update).

In aim/cool, face the player. With sight, expired cooldown, living target, aim error below .14 rad and attack ownership, warn for .5 s, then fire four hitscan rounds .11 s apart. Cooldown is uniform 1.8–3 s. After the first burst, 60% reposition and 40% allow a second burst; after two, reposition. Without sight, reposition after 1.1 s unseen or 3 s aiming. At less than 5 m, reposition after .8 s aiming.

From move, aim or cool, distance below 3 m and expired melee cooldown interrupts ranged ownership: wind up .28 s → lunge .36 s at 9.5 m/s → recover .55 s → reposition. Distance below 1.35 m deals 15 damage once and ends the lunge. Melee cooldown is 3.5 s.

Hound shots aim at player feet + 1 m. Add independent uniform noise to each direction component with half-width `lerp(.12,.045,exposure)*(1+filteredPlayerSpeed*.3)`, then normalize. Range is 200 m; tracer speed 260 m/s is visual. Player speed filtering uses exponential rate 6/s.

Gull attack machine

While engaged, visible, cooldown expired, facing error below .35 rad and target within 42 m, acquire ownership and warn .55 s. Leaving engagement cancels the warning. Fire three bolts .12 s apart, then release ownership and wait uniform 1.9–3.3 s. Bolts travel at 70 m/s, live 1.3 s and deal 7 damage. World obstruction stops them.

Choose lead fraction uniform .2–.85 per burst. Target height is player feet + 1.05 m; horizontal lead = player velocity × (distance / 70) × lead fraction. Component noise half-width is `lerp(.065` multiplied by 1.5 below half gull health. Firing adds .35 m/s recoil opposite the bolt direction.

Motion and encounter rules

Grounding, flight, reinforcement selection and progression

Hound foot solver

Alternate diagonal pairs, choosing the pair with largest predicted horizontal foot error; weight the pair not used last by 1.25. Let J = speed + abs(yaw rate) × .5. Swing duration N = clamp(.27 − J × .026, .14, .27). Trigger at .09 m error while moving (speed > .15 or yaw rate > .25); while idle for over .25 s use .035 m. Predict targets 1.5N seconds ahead using translation and yaw rate. Sample ground below each landing target.

Swing interpolation is smoothstep $t^2(3-2t)$, plus sin(πt) × lift; lift = .065 + min(.09,J × .02). Stance feet stay planted in world space. Solve abduction and two-link leg IK after undoing body pitch/roll/yaw. Effective lower link length subtracts .6 × foot radius; add that amount to foot target Y.

Pitch target = .6 × atan2(rear mean foot Y − front mean foot Y,.8), plus clamp(acceleration × .006,−.07,.07). Roll = .6 × atan2(left mean Y − right mean Y,.55), plus clamp(yawRate × speed × .02,−.12,.12). Body height offset = .45 × (mean foot Y − ground body Y) − min(.05,speed × .011). Height spring stiffness/damping is 160/17, pitch and roll 130/15; add idle breathing sin(time × 1.7 + phase) × .004 m. Antenna springs 60/5 follow acceleration and turning, clamped ±.6 rad.

Gull steering

During combat orbit the player at radius 12–25 m, altitude offset 4–8 m. Every 3–8 s, reverse direction with 60% probability and perturb radius by ±3 m, clamped 13–23 m. Orbit tangential rate is 3.6/radius rad/s, reduced to .9/radius during warning/burst. After 1.5 s unseen, steer toward last known X/Z and last known Y + altitude + 1.

Investigation/search orbit radius 4.5 m at .5 rad/s, speed cap 5.5 m/s, altitude last known Y + min(altitude,5.5). Patrol circles home at .28 rad/s and 3 m/s. Steering acceleration is horizontal error × 2.2 − velocity × 2.4, vertical error × 5 − vertical velocity × 3.6, capped to 11 m/s^2. Combat speed cap is 8 m/s.

Cycle feelers forward, down and up. Forward reach is 2.2 + horizontal speed × .5; obstruction adds normal × 26 × proximity and upward 9 × proximity. Strong proximity over .5 reverses orbit. Body clearance is .62 m horizontally and .35 vertically; collisions reverse the affected velocity component at 25%. Maintain ground clearance 1.1 m. Gimbal follows target exponentially at 9/s; pitch −.5…1.5 rad; gun pitch −.2…1.3 rad, compensating body pitch.

Death removes one random rotor, adds upward velocity 1.5 m/s and spin ±5–8 rad/s. Gravity 9.5 m/s^2; horizontal drag .4/s; spin accelerates 7 rad/s^2. Collision, time over 5 s or Y below −3 ends the fall in a radius-3 m, 30-damage explosion.

Spawn scoring and pressure

For each candidate score uniform(0,1.5), plus 2 + .5 × min(1, (distance−preferred)/30) beyond preferred distance, otherwise minus (preferred−distance)/4. Add 2.5 if hidden from player eye; subtract 3 if used in the last 6 s. Highest score wins. Preferred distance is 32 m for initial patrol and 18 m for waves. Apply X/Z jitter ±1.2 m if airborne or walkable. Air spawns below 2.5 m rise to uniform 6–9 m.

Wave order interleaves hounds into the gull list at index 1, then 3, then 5. Schedule from cut time + 1.6 s with .85 s spacing. Spawn at most one queued enemy per update. At nine living enemies defer due entries 1 s. Failed activation retries in 1.5 s, up to six attempts. A successful arrival alerts enemies within 60 m toward the player.

After the last cut, wait 12 s. When queue empty and fewer than four enemies live, schedule one more after .5 s, 60% gull / 40% hound; allow five extras and reset timer to 7 s. When due but blocked, reset to 4 s and alert survivors within 90 m.

Mission flow and feedback

Title → intro (3.5 s; retry 1.3 s) → playing. Pause freezes the mission clock and simulation. Retry clears all enemies, wrecks, projectiles, cuts and combat statistics and restores full health/ammunition; only retry count persists. Dead presentation desaturates toward .12 and fades toward .55, showing the death menu after 1.5 s. Completion disables movement/damage immediately, fades toward .72 and shows debrief after 3.3 s.

No-shot accuracy is unavailable and contributes zero accuracy points. Ratings map total points 0–5 to Messy, Loud, Steady, Clean, Sharp, Surgical. Kills are displayed but do not affect rating.

Music intensity is 1 during extraction, queued waves or active wave pressure, .6 for recent alerts or enemies within 26 m, otherwise .15. Alerts persist 10 s, damage at least 8 s; wave pressure lasts 30 s but clears early once all enemies and queue are gone after its first 6 s.

Level assembly schedule

Container tiers and elevated route / metres and radians

Containers run along X at yaw zero. Standard 20 ft is 6.06 × 2.438 × 2.59 m (length × width × height); 40 ft is 12.19 × 2.438 × 2.59 m; high cube (HC) is 12.19 × 2.438 × 2.9 m. Entries below are bottom to top. X/Z are stack centers; base Y is zero. Tier Y equals the sum of lower tier heights. Jitter is a multiplier: upper X offsets add uniform ±.05 × jitter, Z ±.04 × jitter; yaw adds uniform ±.022 × jitter radians (× .6 above tier zero). Authored dx/dz offsets add in world X/Z. Door numbers are opening amounts; flip reverses orientation.

X / Z	Yaw	Jitter	Tiers: size, color, options
−19.45 / −5.12	0	1	(20, "green") → (20, "grey") → (20, "orange")
−13.3 / −5.12	0	1	(20, "white", (reefer: true, flip: true))
3.7 / −5.12	0	1	(20, "blue", (doors: 0.8, flip: true)) → (20, "white")
13.53 / −5.12	0	1	(20, "teal") → (20, "grey")
−19.45 / 2.08	0	1	(20, "blue")
−13.35 / 2.08	0	1	(20, "red")
0.8 / 2.08	0	1	(40, "orange") → (40, "teal") → (20, "blue", (dx: 3))
12.3 / 2.08	0	1	(20, "red") → (20, "yellow")
−21.1 / 9.92	0	0.3	(20, "yellow")
−13.6 / 9.92	0	0.3	(20, "green") → (20, "blue") → (20, "orange")
−21.6 / 12.42	0	0.3	(20, "red") → (20, "white") → (20, "teal")
−15.4 / 12.42	0	0.3	(20, "grey", (doors: 0.6))
4.43 / 12.42	0	0	(20, "blue")
13.53 / 12.42	0	0	(20, "green")
−16.4 / 20.12	0	1	(40, "grey") → (40, "green")
−2.25 / 20.12	0	1	(20, "red")
3.85 / 20.12	0	1	(20, "white") → (20, "blue")
13.53 / 20.12	0	1	(20, "orange") → (20, "teal") → (20, "red")
−16.4 / 22.62	0	1	("hc", "teal") → (40, "yellow") → (40, "blue")
0.8 / 22.62	0	1	(40, "green") → ("hc", "orange")
13.53 / 22.62	0	1	(20, "grey", (doors: 0.4))
−22.6 / −19.2	1.571	1	(40, "red") → (40, "white")
−30.2 / −18.4	0.35	0	(20, "yellow", (doors: 0.5))
−31.2 / 24.7	0	1	(40, "teal") → (40, "orange")
23.2 / −28.6	0	1	(20, "white", (reefer: true)) → (20, "white", (reefer: true))
23 / −25.95	0	1	(20, "grey", (reefer: true))

Stairs at (−.1,0,12.42), yaw 0, rise to the blue container roof; catwalk at (8.98,2.54,12.42) bridges blue and green containers. Four support rods stand at X 7.6/10.36 and Z 11.86/12.98 up to Y 2.5. Roof cover is placed at Y 2.59. Keep these surfaces continuous and traversable.

Authored prop placement

Yard, gate approach and elevated cover

Each row gives builder, X/Z, yaw, options, base Y. Displayed measurements are rounded to three decimal places. Unspecified elevation is ground level. Use catalogue dimensions and the named executable builder for visual construction and detailed collision. Props marked noCol form part of the forklift load, with its separate wood collision envelope.

Asset	X / Z	Yaw	Options	Y
sandbags	15.95 / 12.42	1.571	(seed: 61)	2.59
crate_wood	14.3 / 11.75	0.1	(seed: 62)	2.59
sandbags	2.3 / 11.75	0	(seed: 63)	2.59
light_mast	−23.4 / 0.5	0	(seed: -70)	0
light_mast	−23.4 / 21.4	0	(seed: -49)	0
light_mast	17 / −8.7	0	(seed: 42)	0
light_mast	17 / 18.6	0	(seed: 70)	0
light_mast	−20.6 / −11.6	0	(seed: -73)	0
barrel	−19.6 / 3.85	0	(seed: 1, color: "blue")	0
barrel	−18.9 / 3.95	0.7	(seed: 2, color: "blue")	0
pallet_stack	−17.4 / 4	0.08	(seed: 3)	0
crate_stack	−14.3 / 8.05	0.04	(seed: 4)	0
ibc_tote	−12.95 / 7.95	−0.05	(seed: 5)	0
forklift	−8.6 / 4.5	−0.22	(seed: 6, forksUp: 0.3)	0
pallet	−7.214 / 4.81	−0.22	(seed: 7)	0.535
crate_wood	−7.214 / 4.81	−0.17	(seed: 8)	0.679
jersey_barrier	−2.9 / 5.3	0.12	(seed: 11)	0
jersey_barrier	5.6 / 5	−0.1	(seed: 12)	0
sandbags	−1.7 / 9.7	0.5	(seed: 13)	0
ibc_tote	5.9 / 8.9	0.1	(seed: 14)	0
ibc_tote	5.95 / 10.05	−0.05	(seed: 15)	0
barrel	4.9 / 9.7	0	(seed: 16, color: "red")	0
crate_stack	−3.9 / 10.5	0.1	(seed: 17)	0
crate_wood	−3.8 / 9.35	−0.2	(seed: 18)	0
pallet_stack	−4.4 / 11.8	0.02	(seed: 19)	0
cable_reel	4.2 / 10.5	0.3	(seed: 20)	0
tarp_cargo	12.9 / 10	0.04	(seed: 21)	0
jersey_barrier	11.6 / 3.95	0.03	(seed: 81)	0
ibc_tote	15.2 / 10.2	0.1	(seed: 82)	0
traffic_cone	0.2 / 6.3	0	(seed: 33)	0
traffic_cone	2.9 / 6.4	0	(seed: 58)	0
traffic_cone	2.9 / 8.8	0	(seed: 70)	0
traffic_cone	0.1 / 8.7	0	(seed: 44)	0
sign_post	−4.8 / 3.72	1.571	(seed: 22, sign: "hazard_voltage")	0
tarp_cargo	−13.5 / −3.1	0.05	(seed: 23)	0
pallet_stack	−20.6 / −0.1	0.2	(seed: 24)	0
tire_stack	−9.6 / 0.2	0	(seed: 25)	0
barrel	4.5 / 0.35	0	(seed: 26, color: "yellow")	0
barrel	5.2 / 0.3	0.4	(seed: 27)	0
jersey_barrier	12.2 / 0.4	0.03	(seed: 28)	0
crate_wood	11.2 / −3.3	0.3	(seed: 29)	0
cardboard_boxes	−2 / 0.3	0.2	(seed: 30)	0
jersey_barrier	−0.9 / −7.5	0.1	(seed: 31)	0
sandbags	−6.7 / −7.8	−0.1	(seed: 32)	0

Asset	X / Z	Yaw	Options	Y
dumpster	−12.3 / −7.05	0	(seed: 33)	0
gas_cylinders	9.6 / −10.45	0	(seed: 34)	0
pallet_stack	1.2 / −9.9	0.1	(seed: 35)	0
barrel	2.2 / −10.3	0	(seed: 36, color: "green")	0
cable_reel	−18.2 / −8	0.2	(seed: 37)	0
traffic_cone	−6.5 / −9.9	0	(seed: 38)	0
traffic_cone	−0.5 / −9.9	0	(seed: 39)	0
pipe_stack	−16 / 14.4	0	(seed: 41)	0
crate_stack	−10.95 / 18.25	0.06	(seed: 42)	0
barrel	−3 / 18.35	0	(seed: 43, color: "blue")	0
dumpster	4.2 / 14.55	0.02	(seed: 44)	0
tire_stack	10.95 / 18.3	0	(seed: 45)	0
ibc_tote	12 / 18.25	0	(seed: 46)	0
pallet_stack	−12.3 / 18.2	0.1	(seed: 47)	0
tarp_cargo	−20.5 / 18.05	0.03	(seed: 48)	0
jersey_barrier	0.6 / 15.5	0.12	(seed: 49)	0
tire_stack	−8 / 26.9	0	(seed: 51)	0
pallet_stack	8.7 / 27	0.3	(seed: 52)	0
crate_wood	−2 / 27	0.2	(seed: 53)	0
barrel	11.2 / 27.1	0	(seed: 54)	0
cardboard_boxes	−21 / 26.9	0.4	(seed: 55)	0
pipe_stack	−30.5 / −27	0.03	(seed: 56)	0
pipe_stack	−30.2 / −25.4	−0.02	(seed: 57)	0
dumpster	−17 / −14	1.571	(seed: 58)	0
tarp_cargo	−27.5 / −11.6	0.3	(seed: 59)	0
tarp_cargo	−29.9 / −11	−0.2	(seed: 60)	0
cable_reel	−34.6 / −22.5	1	(seed: 64)	0
tire_stack	−25.4 / −26.9	0	(seed: 65)	0
gas_cylinders	−16.9 / −20.4	1.571	(seed: 66)	0
crate_stack	−34.8 / −13.5	0.2	(seed: 67)	0
barrel	−33.6 / −13	0	(seed: 68)	0
generator	−34.2 / 14.2	0.15	(seed: 71)	0
floodlight_stand	−31.4 / 12.7	0.55	(seed: 72)	0
tire_stack	−36.6 / 17.2	0	(seed: 73)	0
tire_stack	−36.5 / 18.3	0.6	(seed: 74)	0
dumpster	−36.4 / 21	1.571	(seed: 75)	0
pallet_stack	−27.8 / 16.6	0.4	(seed: 76)	0
jerry_can	−33 / 15.2	0.4	(seed: 77)	0
jerry_can	−33.3 / 15.35	1.9	(seed: 78)	0
crate_wood	−28.6 / 20.6	0.1	(seed: 79)	0
ibc_tote	−26.6 / 21.2	0	(seed: 80)	0
pallet_stack	0 / −26.9	0.1	(seed: 3)	0.03
cardboard_boxes	−0.05 / −26.9	0.5	(seed: 4)	1.18
crate_stack	2.9 / −15.9	0.12	(seed: 11)	0.03
crate_wood	4.15 / −16.3	−0.25	(seed: 12)	0.03
pallet_stack	2 / −17.3	1.45	(seed: 13)	0.03
ibc_tote	6.9 / −19.6	0.1	(seed: 14)	0.03
barrel	7.9 / −18.5	0	(seed: 15, color: "blue")	0.03
barrel	6 / −18.4	0.6	(seed: 16, color: "red")	0.03
ibc_tote	−7.6 / −16.4	−0.08	(seed: 17)	0.03
barrel	−8.6 / −15.3	0	(seed: 18)	0.03
barrel	−7.9 / −15	0.9	(seed: 19, color: "yellow")	0.03
sandbags	10.2 / −16.4	1.621	(seed: 20)	0.03
forklift	8.3 / −13.9	2.75	(seed: 21)	0.03
toolbox_cart	6 / −12.2	0.15	(seed: 22)	0.03

Asset	X / Z	Yaw	Options	Y
gas_cylinders	10.9 / −11.9	0	(seed: 23)	0.03
tarp_cargo	−13.2 / −13.2	0.12	(seed: 24)	0.03
tire_stack	−15 / −15.6	0	(seed: 25)	0.03
tire_stack	−14.9 / −16.6	0.4	(seed: 26)	0.03
cardboard_boxes	−14.6 / −26.6	0.2	(seed: 27)	0.03
crate_wood	−12.4 / −26.9	−0.1	(seed: 28)	0.03
barrel	−10.9 / −27	0	(seed: 29, color: "green")	0.03
crate_wood	−14.7 / −24.6	0.3	(seed: 30)	0.03
jerry_can	−9.3 / −12	0.4	(seed: 31)	0.03
jerry_can	−9.7 / −12.1	1.2	(seed: 32)	0.03
pallet	10.6 / −13.6	0.2	(seed: 33)	0.03
cardboard_boxes	10.6 / −13.6	0.3	(seed: 34)	0.17
traffic_cone	−6.6 / −12.2	0	(seed: 35)	0.03
traffic_cone	−0.4 / −12.3	0	(seed: 36)	0.03
jersey_barrier	20.5 / −4.7	1.691	(seed: 31)	0
jersey_barrier	25.3 / 3.1	−0.2	(seed: 32)	0
sandbags	20.6 / 3	1.25	(seed: 33)	0
barrel	26.5 / −1.7	0	(seed: 34, color: "red")	0
barrel	26.9 / −1.05	0.5	(seed: 35)	0
traffic_cone	22 / −1.9	0	(seed: 36)	0
traffic_cone	24.2 / 1.7	0	(seed: 37)	0
generator	15 / −26.6	0.06	(seed: 41)	0
floodlight_stand	16.4 / −23.6	0.3	(seed: 42)	0
tire_stack	24.1 / −9.5	0	(seed: 43)	0
tire_stack	25.05 / −9.6	0.5	(seed: 44)	0
cable_reel	14.3 / −12.9	0.4	(seed: 45)	0
cable_reel	15.6 / −14.3	1.9	(seed: 46)	0
pallet_stack	20.3 / −9.3	0.1	(seed: 47)	0
crate_wood	19.9 / −22.6	0.3	(seed: 48)	0
barrel	12.75 / −23.3	0	(seed: 49, color: "blue")	0
barrel	12.9 / −24.1	0	(seed: 50, color: "blue")	0
light_mast	26.6 / −28.7	0	(seed: 51)	0
light_mast	26.6 / 26.7	0	(seed: 52)	0
crate_stack	26.4 / 23.6	0.2	(seed: 53)	0
ibc_tote	22.6 / 26.3	0.05	(seed: 54)	0
pallet_stack	24 / 26.2	0.2	(seed: 55)	0
barrel	22.6 / 20.3	0	(seed: 56, color: "yellow")	0
barrel	23.4 / 20.4	0	(seed: 57)	0
tarp_cargo	20.3 / 12.6	1.571	(seed: 59)	0
traffic_cone	28.2 / 18.2	0	(seed: 60)	0
traffic_cone	28.3 / 21.8	0	(seed: 61)	0
shelf_rack	−8.4 / −27.13	0	(:)	0.03
shelf_rack	−5.6 / −27.13	0	(:)	0.03
shelf_rack	−2.8 / −27.13	0	(:)	0.03
shelf_rack	2.8 / −27.13	0	(:)	0.03
shelf_rack	5.6 / −27.13	0	(:)	0.03
shelf_rack	8.4 / −27.13	0	(:)	0.03
shelf_rack	2.8 / −23.45	0	(:)	0.03
shelf_rack	5.6 / −23.45	0	(:)	0.03
shelf_rack	8.4 / −23.45	0	(:)	0.03
fence_panel	−36.3 / −30	0	(seed: 100, posts: "left")	0
fence_panel	−32.9 / −30	0	(seed: 101, posts: "left")	0
fence_panel	−29.5 / −30	0	(seed: 102, posts: "left")	0
fence_panel	−26.1 / −30	0	(seed: 103, posts: "left")	0
fence_panel	−22.7 / −30	0	(seed: 104, posts: "left")	0

Asset	X / Z	Yaw	Options	Y
fence_panel	−19.3 / −30	0	(seed: 105, posts: "left")	0
fence_panel	−15.9 / −30	0	(seed: 106, posts: "left")	0
fence_panel	−12.5 / −30	0	(seed: 107, posts: "left")	0
fence_panel	−9.1 / −30	0	(seed: 108, posts: "left")	0
fence_panel	−5.7 / −30	0	(seed: 109, posts: "left")	0
fence_panel	−2.3 / −30	0	(seed: 110, posts: "left")	0
fence_panel	1.1 / −30	0	(seed: 111, posts: "left")	0
fence_panel	4.5 / −30	0	(seed: 112, posts: "left")	0
fence_panel	7.9 / −30	0	(seed: 113, posts: "left")	0
fence_panel	11.3 / −30	0	(seed: 114, posts: "left")	0
fence_panel	14.7 / −30	0	(seed: 115, posts: "left")	0
fence_panel	18.1 / −30	0	(seed: 116, posts: "left")	0
fence_panel	21.5 / −30	0	(seed: 117, posts: "left")	0
fence_panel	24.9 / −30	0	(seed: 118, posts: "left")	0
fence_panel	28.3 / −30	0	(seed: 119, posts: "both")	0
fence_panel	−36.3 / 28	0	(seed: 200, posts: "left")	0
fence_panel	−32.9 / 28	0	(seed: 201, posts: "left")	0
fence_panel	−29.5 / 28	0	(seed: 202, posts: "left")	0
fence_panel	−26.1 / 28	0	(seed: 203, posts: "left")	0
fence_panel	−22.7 / 28	0	(seed: 204, posts: "left")	0
fence_panel	−19.3 / 28	0	(seed: 205, posts: "left")	0
fence_panel	−15.9 / 28	0	(seed: 206, posts: "left")	0
fence_panel	−12.5 / 28	0	(seed: 207, posts: "left")	0
fence_panel	−9.1 / 28	0	(seed: 208, posts: "left")	0
fence_panel	−5.7 / 28	0	(seed: 209, posts: "left")	0
fence_panel	−2.3 / 28	0	(seed: 210, posts: "left")	0
fence_panel	1.1 / 28	0	(seed: 211, posts: "left")	0
fence_panel	4.5 / 28	0	(seed: 212, posts: "left")	0
fence_panel	7.9 / 28	0	(seed: 213, posts: "left")	0
fence_panel	11.3 / 28	0	(seed: 214, posts: "left")	0
fence_panel	14.7 / 28	0	(seed: 215, posts: "left")	0
fence_panel	18.1 / 28	0	(seed: 216, posts: "left")	0
fence_panel	21.5 / 28	0	(seed: 217, posts: "left")	0
fence_panel	24.9 / 28	0	(seed: 218, posts: "left")	0
fence_panel	28.3 / 28	0	(seed: 219, posts: "both")	0
fence_panel	−38 / −28.4	1.571	(seed: 300, posts: "left")	0
fence_panel	−38 / −25.2	1.571	(seed: 301, posts: "left")	0
fence_panel	−38 / −22	1.571	(seed: 302, posts: "left")	0
fence_panel	−38 / −18.8	1.571	(seed: 303, posts: "left")	0
fence_panel	−38 / −15.6	1.571	(seed: 304, posts: "left")	0
fence_panel	−38 / −12.4	1.571	(seed: 305, posts: "left")	0
fence_panel	−38 / −9.2	1.571	(seed: 306, posts: "left")	0
fence_panel	−38 / −6	1.571	(seed: 307, posts: "left")	0
fence_panel	−38 / −2.8	1.571	(seed: 308, posts: "left")	0
fence_panel	−38 / 0.4	1.571	(seed: 309, posts: "both")	0
fence_panel	−38 / 9.667	1.571	(seed: 400, posts: "left")	0
fence_panel	−38 / 13	1.571	(seed: 401, posts: "left")	0
fence_panel	−38 / 16.333	1.571	(seed: 402, posts: "left")	0
fence_panel	−38 / 19.667	1.571	(seed: 403, posts: "left")	0
fence_panel	−38 / 23	1.571	(seed: 404, posts: "left")	0
fence_panel	−38 / 26.333	1.571	(seed: 405, posts: "both")	0
guard_booth	−35.2 / 0.35	0	(seed: 1)	0
gate_arm	−36.3 / 5	1.571	(seed: 2)	0
bollard_post	−37 / 1.75	0	(seed: -395)	0
bollard_post	−34.4 / 2.3	0	(seed: -362)	0

Asset	X / Z	Yaw	Options	Y
bollard_post	−37 / 8.3	0	(seed: -349)	0
bollard_post	−35.6 / 8.9	0	(seed: -329)	0
bollard_post	−34.4 / 9.9	0	(seed: -309)	0
bollard_post	−33.5 / −1.1	0	(seed: -376)	0
jersey_barrier	−34.7 / 3.75	1.651	(seed: 3)	0
jersey_barrier	−33.3 / 8.3	1.511	(seed: 4)	0
jersey_barrier	−30.6 / 10.4	0.03	(seed: 5)	0
jersey_barrier	−27.3 / 1.45	0.05	(seed: 6)	0
jersey_barrier	−32.2 / 1.45	−0.04	(seed: 7)	0
sign_post	−33.2 / 9.6	1.571	(seed: 8, sign: "no_entry")	0
sign_post	−25.3 / 1.1	1.571	(seed: 9, sign: "berth4")	0
sign_post	−36.6 / −1.6	1.571	(seed: 10, sign: "hazard_general")	0
traffic_cone	−34.6 / 5.6	0	(seed: -156)	0
traffic_cone	−28.3 / 2	0	(seed: -135)	0
traffic_cone	−33.2 / 6.5	0	(seed: -146)	0
traffic_cone	−26.6 / 9.8	0	(seed: -104)	0
traffic_cone	−31.8 / 9.9	0	(seed: -129)	0
light_mast	−27.6 / −0.9	0	(seed: 11)	0
light_mast	−36.3 / 12.4	0	(seed: 12)	0
barrel	−33.9 / 2	0	(seed: 13, color: "yellow")	0
pallet_stack	−36.4 / 10.3	0.2	(seed: 14)	0

Spawn coordinate schedule

Authored candidates / ground positions are repaired to walkable navigation

Kind	X / Y / Z	Zone
ground	−19 / 0 / −1.4	yard
ground	3.5 / 0 / −1.6	yard
ground	14.2 / 0 / 16.4	yard
ground	−3.2 / 0 / 16.6	yard
ground	−16 / 0 / 26	yard
ground	6 / 0 / 26	yard
ground	4.2 / 0 / −5.12	yard
ground	−7.8 / 0 / 21.4	yard
ground	−11.3 / 0 / 12.5	yard
ground	17.5 / 0 / 22.2	yard
air	10 / 11 / 23.5	yard
air	2 / 11 / 21.5	yard
air	13 / 9 / 11	yard
air	−4 / 12.5 / −4.5	yard
air	9 / 10 / 1.5	yard
air	−10 / 12 / 24.5	yard
ground	−12.4 / 0 / −24.8	shed
ground	9.6 / 0 / −25.3	shed
ground	−13.4 / 0 / −19.6	shed
ground	4.6 / 0 / −13.8	shed
ground	−21 / 0 / −27.2	shed
ground	−18.6 / 0 / −22	shed
ground	13.6 / 0 / −24	shed
ground	−26.6 / 0 / −15	shed
air	−8 / 5.4 / −20.5	shed
air	5 / 5.2 / −18	shed
air	−20 / 10.5 / −20	shed
air	14.5 / 10 / −22	shed
air	−2 / 12.5 / −8.5	shed
ground	27.7 / 0 / −24.2	quay
ground	18.3 / 0 / −25.4	quay
ground	27.2 / 0 / −12.6	quay
ground	20.2 / 0 / 14	quay
ground	23.4 / 0 / 23.2	quay
ground	18 / 0 / 24.4	quay
ground	27.4 / 0 / 11.6	quay
air	36 / 9 / 4	quay
air	38 / 11 / −6	quay
air	35 / 8.5 / 16	quay
air	24 / 12.5 / 10.5	quay
air	33.5 / 10 / 24.5	quay
air	22 / 13 / −14	quay

Boundary and quay

Ground yard bounds are X −40…40, Z −30…30; clamp player centers .4 m inside. Sea level is −1.6 m. Quay lip is X 30, Z −31…29, from Y −.7 to .028, with a .65 m concrete width. The sea-facing wall extends to −4.2 m. Yellow edge paint, mooring bollards and tire fenders establish the hazardous edge.

The ladder is at X 29.88, Z 20, with side rails Z 19.74/20.26 down to −3.9 m and rungs every .3 m. Its .84 m opening spans Z 19.58…20.42. Yellow safety posts at X 29.18 stand at Z 17.4,19.1,20.9,22.6 with chains between the outer pairs, leaving the middle passage open. Extraction is the ground-side radius around (29,0,20), without a climb requirement.

Bollards stand X 29.2 at Z −22,−12,−2,9,16.4,25.6. Crane rails run along Z −29.6…27.6 at X 18.5 and 28.5. Preserve their small .034 m raised steel profiles rather than treating them as high walls.

Complete movement tuning

Camera, strides and bodily feedback

Exponential following at rate r means new = target + (old − target) × exp(−r × elapsed seconds). Spring stiffness/damping uses acceleration = stiffness × (target − position) − damping × velocity. Angles below are radians except FOV in degrees. Strides are metres per cycle; movement speeds are m/s, acceleration and gravity m/s².

Parameter	Value
walk	3.8
sprint	6.4
crouch	1.9
ads	2.4
accelGround	34
decelGround	26
accelAir	9
gravity	21
jumpApex	1
coyote	0.12
jumpBuffer	0.13
standHeight	1.8
crouchHeight	1.15
eyeRate	11
stride	(walk: 1.55, sprint: 2.1, crouch: 1.05)
bobY	0.024
bobX	0.013
bobSprint	1.9
bobAds	0.22
rollStrafe	0.02
landDip	0.36
landDipMax	0.2
fovSprint	6
fovAds	52
adsLook	0.6
regenDelay	4
regenRate	25
recoilKeep	0.35
recoilApply	34
recoilRecover	6.5
recoilHold	0.09
shakeDecay	1.7
shakePitch	0.03
shakeYaw	0.03
shakeRoll	0.045
pitchLimit	1.5

Look and landing

Camera bob amplitudes are .024 m vertical, .013 m horizontal, multiplied by 1.9 sprinting or .22 while aiming. Keep the camera inside the capsule and blend eye height at rate 11/s. Jump launch speed is sqrt(2 × 21 × 1) ≈ 6.481 m/s. Terminal downward speed is 40 m/s. Crouch cannot end under a ceiling. Landing shake starts above 7.5 m/s at min(.5, (impact−7.5) × .06); the landing spring receives a downward velocity impulse min(impact × .36,3.2).

Combat volumes and scene direction

Hit feedback, warehouse access and presentation anchors

Damage resolution

Player shots stop at the nearest world surface or enemy volume; there is no penetration or damage falloff. Hound optics use a .155 m radius weak-point sphere at the animated head: damage multiplier 2.2 (57.2 per PAC-9 hit). Body targeting uses two longitudinal rounded segments, endpoints −.45 to +.47 m along body forward: upper radius .27 m at body center + .03 m; lower radius .25 m at 45% of body height. Use the closest intersected volume. Gull body is an ellipsoid centered .03 m below its root, horizontal radius .5 m and vertical radius .5/2.3 m; no weak point.

Hound bullets test the player around feet +1 m, with horizontal hit tolerance .38 m and vertical center .95 m ±.95; world obstruction wins. Gull bolts sweep against a .36 m radius segment from feet +.35 to feet +1.5 m (crouching +1 m). Distinguish ordinary, weak-point and lethal hitmarkers; count one hit per damaging player shot.

Explosions require line of sight and linear falloff. Enemies receive base damage × (1−distance/radius), measured to root position, with visibility tested at root +.5 m. Player damage is half that base, measured and visibility-tested at feet +1 m. Thus a gull crash has maximum player damage 15, despite its 30 base damage against enemies. Explosions can cause enemy chain destruction.

Warehouse and perimeter

Warehouse walls are .25 m thick. Main south-facing opening at Z −11 spans X −6…−1, height 5 m. East opening at X 12 spans Z −22…−17, height 4.5 m. Preserve clear approaches to both and the central relay (−2,.03,−19.5). Lower wall sections are concrete to 1.2 m, corrugated cladding above; pitched corrugated roof rises from 7 m eaves to 9.5 m at Z −19.5. Epoxy floor is Y .03. Shelf rows and stored goods leave circulation around relay and doors.

Chain-link perimeter follows north Z −30 and south Z 28 from X −38 to 30; west fence X −38 has its gate opening at Z 2…8. Fence physical height is 2.4 m with a 7 m player barrier to prevent climbing out. The open sliding gate rests centered Z 11.3, X −37.72. Guard booth at (−35.2,0,.35), raised gate arm at (−36.3,0,5) and concrete barriers make deployment read as a checkpoint. The west gate also has a player-only blocker across its opening: the mission remains inside the berth.

Default controls

WASD move, mouse look, left mouse/Enter fire, right mouse aim, Shift sprint, C or Control crouch, Space jump, R reload, hold E interact, Escape/P pause. Enter begins deployment from title. Arrow keys offer turning. Keep displayed keyboard, gamepad and touch prompts aligned with their active device. Bound melee/alternate inputs have no additional player combat action; PAC-9 is the mission weapon.

Palette and lighting

Use low warm sun against cold storm shadows. Sun direction is normalized (.916,.225,.331), linear RGB (1,.72,.42), intensity 9. Sky zenith is (.055,.085,.13), sunward horizon (.95,.55,.28), opposite horizon (.16,.22,.3); fog (.42,.47,.52). These are lighting values, not flat display colors. Preserve wet specular asphalt, pale corrugated metal, rust streaks, orange/teal container contrast and dark seawater.

Relay lights, gull searchlights and robot lenses carry gameplay contrast through rain. Warehouse shelter suppresses rain and changes reverberation; outside shots have an open tail, interior shots a denser reflected tail. Rotor and servo pitch follow motion; warning sounds begin with visible attack telegraphs. Magazine-out, insertion and bolt audio follow the exact reload milestones. Radio announces deployment, first contact, relay progress, final extraction instruction, low health once, rescue and completion; it must not obscure warning cues.

Flight clearance detail

Downward feelers reach 2.4 m and add upward avoidance (2.4−hit distance) × 10. Upward feelers reach 4 m and set ceiling target to hit Y −1.3 m; refresh .35 s on that feeler cycle. Avoidance decays exponentially at 6/s. Below half health, oscillatory force amplitudes 5/3/5 on X/Y/Z destabilize flight while smoke runs every .16 s.

Tactical positioning

Hound firing positions and player movement qualifications

Choosing firing positions

Consider the authored cover positions below with score bias −2; seven random walkable positions within 18 m of the player with bias 0; and five positions 9–17 m from the player, at angles within ±1.3 rad of the current player-to-hound bearing, with bias +1. Reject unwalkable positions and distances outside 8–18 m.

Minimize .9 × travel distance + bias + uniform(0,3). For each other engaged hound add 16 if within 3 m of its current/intended firing position; add `14*(directionDot-.86)/.14` when both attack directions from player have dot > .86. Add 22 if the candidate has no sight from Y +.95 m to player eye. If none qualify, stay at the current position.

Navigate around world cover. Stop within .45 m of the goal. For yaw error above 1.1 rad stop to turn; otherwise speed cap is min(requestedSpeed × (1−yawError × .5), .4 + goalDistance × 2.6). Maximum turn rate is 2.5 rad/s, doubled for lunges, following desired turn velocity at 10/s. Patrol waits 1.5–4.5 s between destinations. Investigation approaches last known position until within 2.2 m; use 2 m/s within 8 m, otherwise 4.5 m/s, then scan yaw with sin(scanTime × .8) × .9 rad.

Movement qualifications

Sprint requires forward input > .55, lateral magnitude < .6, no crouch, aim blend < .35 and no firing. Sprint presentation begins grounded above 60% walk speed and persists in air if already sprinting. Aim interpolates movement speed toward min(current speed,2.4); backwards input scales speed toward .82 at full reverse. Ground input acceleration is 34, braking 26; airborne acceleration 9 and no-input drift deceleration .6 m/s². Horizontal velocity approaches the desired vector by at most acceleration × elapsed seconds.

Authored cover schedule

All values are X/Y/Z metres. Repair blocked ground-level points by testing radii .5,1,1.5,2 m in eight directions at angle increments .7854 rad; take the first walkable position. Apply the same repair to ground spawn candidates. Elevated points above .2 m are retained.

X / Y / Z	X / Y / Z
−33.9 / 0 / 3.8	−27.3 / 0 / 2.3
−19.2 / 0 / 4.7	−17.4 / 0 / 4.9
−12.5 / 0 / 7.2	−9.3 / 0 / 6.4
−6.3 / 0 / 7.3	−2.9 / 0 / 6.1
−2.9 / 0 / 4.5	5.6 / 0 / 5.8
5.6 / 0 / 4.2	−2.2 / 0 / 10.4
5 / 0 / 8.9	−4.8 / 0 / 10.6
7.5 / 0 / 7.7	9.2 / 0 / 4.4
−1 / 0 / 13.6	−13.5 / 0 / −1.5
−9.6 / 0 / −0.7	7.8 / 0 / −1.5
−4.2 / 0 / −2.2	−0.9 / 0 / −6.7
−6.7 / 0 / −7	1.2 / 0 / −8.9
−8.1 / 0 / 15.4	0.6 / 0 / 16.9
4.2 / 0 / 15.7	−16 / 0 / 15.6
12 / 0 / 17.2	2.9 / 0 / −17
4.2 / 0 / −15.3	6.9 / 0 / −20.6
−7.6 / 0 / −17.4	−8.3 / 0 / −14.3
9.6 / 0 / −21.9	−13.2 / 0 / −14.4
−11.6 / 0 / −18.6	0 / 0 / −25.8
−12.4 / 0 / −25.8	21.4 / 0 / −4.7
25.3 / 0 / 3.9	25.7 / 0 / −4.4
20.2 / 0 / −15.4	26.4 / 0 / −15.4
20.6 / 0 / 14.4	26.8 / 0 / 14.4
20.6 / 0 / 21.2	27.4 / 0 / 22.4
19.6 / 0 / −21.4	27 / 0 / −16.2
25.6 / 0 / 6	13.5 / 2.59 / 12.4
4.4 / 2.59 / 12.4	12.9 / 0 / 8.8
11.6 / 0 / 4.9	